

Manufacturing of Sterile Pharmaceuticals

Premises of parenteral manufacturing



- Examples:-
 - Parenteral,
 - Ophthalmic, and
 - Irrigating preparations

Advantages and disadvantages of parenteral formulations

Advantages

- An immediate physiological response may be achieved (usually by the IV route). This is important in acute medical situations, e.g. cardiac arrest, anaphylactic shock, asthma.
- Parenteral formulations are essential for drugs that offer poor bioavailability or those that are rapidly degraded within the gastrointestinal tract (e.g. insulin and other peptides).
- They offer a method to administer drugs to patients who are unconscious or uncooperative or for patients with nausea and vomiting (and additionally dysphagia).
- Administration of insulin, which, in the absence of complications (e.g. ketoacidosis), is performed exclusively by patients.

■ Local effects may be achieved using parenteral formulations, e.g. local anaesthesia.

■ Parenteral formulations provide a means by which serious imbalances in electrolytes may be corrected (using infusion solutions).

■ Parenteral formulations may be readily formulated to offer a wide range of drug release profiles, including:

- rapidly acting formulations (generally drug solutions that are administered IV)

- long-acting formulations (generally drug suspensions, or solutions in which the drug is precipitated out of solution at the injection site, administered by the IM or SC routes).

Examples of these include intermediate/long-acting insulin formulations and steroid injections.

■ In patients who cannot consume food, total parenteral nutrition offers a means by which nutrition may be provided using specially formulated solutions that are infused into the patient.

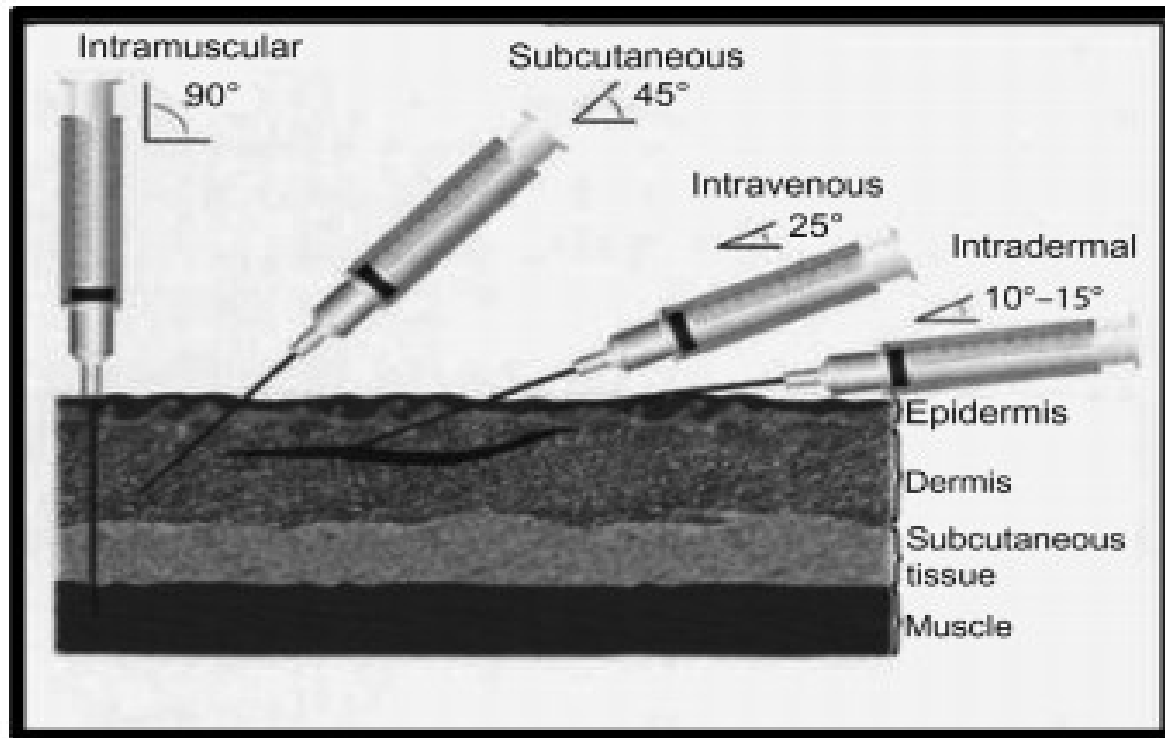
Disadvantages

- The manufacturing process is more complicated than for other formulations due to the requirement for aseptic technique. The level of training of staff involved in the manufacture of parenteral formulations is high and often specialist equipment is required to ensure that the finished product specification is achieved.
- Skill of administration is required to ensure that the dosage form is administered by the correct route. If a parenteral suspension, which is designed for administration by the IM or SC route, is incorrectly administered by the IV route, a pulmonary microcapillary blockage may occur leading to a blockage in the flow of blood at that site.
- Parenteral formulations are associated with pain on administration.
- If the patient is allergic to the formulation (the therapeutic agents and/or the excipients), parenteral administration will result in both rapid and intense allergic reactions.
- It is difficult to reverse the effects of drugs that have been administered parenterally, even immediately after administration. This is not strictly the case with other routes of administration, e.g. oral, transdermal.

PARENTERAL ROUTES OF ADMINISTRATION

Drugs may be injected into almost any organ or area of the body, including the joints (intraarticular), joint fluid area (intrasynovial), spinal column (intraspinal), spinal fluid (intrathecal), arteries (intra-arterial), and in an emergency, even the heart (intracardiac).

However, most injections go into a vein (intravenous, IV), into a muscle (intramuscular, IM), into the skin (intradermal, ID; intracutaneous), or under the skin (subcutaneous, SC; sub-Q, SQ; hypodermic, hypo)



Introduction

- Parenteral products-for use by injection under or through one or more layers of the skin or mucous membranes
- They are most frequently
 - ➡ solutions
 - ➡ suspensions
 - ➡ but may even be solid pellets for tissue implantation
- They must be free from
 - ◆ microbial contamination and
 - ◆ toxic components as well as
 - ◆ possess an exceptionally high level of purity

OFFICIAL TYPES OF INJECTIONS

According to the USP, injectable materials are separated into five general types. These may contain buffers, preservatives, and other added substances.

1. *Injection*: Liquid preparations that are drug substances or solutions thereof (e.g., Insulin Injection, USP).
2. *For injection*: Dry solids that, upon addition of suitable vehicles, yield solutions conforming in all respects to the requirements for injections (e.g., Cefuroxime for injection, USP).
3. *Injectable emulsion*: Liquid preparation of drug substance dissolved or dispersed in a suitable emulsion medium (e.g., Propofol, USP).
4. *Injectable suspension*: Liquid preparation of solid suspended in a suitable liquid medium (e.g., Methylprednisolone Acetate Suspension, USP).
5. *For injectable suspension*: Dry solid that, upon addition of suitable vehicle, yields preparation conforming in all respects to the requirements for *injectable suspensions* (e.g., Imipenem and Cilastatin for injectable suspension, USP).

SOME INJECTIONS IN OIL

INJECTION	OIL	CATEGORY
Dimercaprol	Peanut	Antidote to arsenic, gold, and mercury poisoning
Estradiol cypionate	Cottonseed	Estrogen
Estradiol valerate	Sesame or castor	Estrogen
Fluphenazine decanoate	Sesame	Antipsychotic
Fluphenazine enanthate	Sesame	Antipsychotic
Hydroxyprogesterone caproate	Castor	Progestin
Progesterone in oil	Sesame or peanut	Progestin
Testosterone cypionate	Cottonseed	Androgen
Testosterone cypionate and estradiol cypionate	Cottonseed	Androgen and estrogen
Testosterone enanthate	Sesame	Androgen
Testosterone enanthate and estradiol valerate	Sesame	Androgen and estrogen

Introduction

- In the preparation of these products
⇒all components and all processes must be selected and designed to eliminate contamination of all types, whether of
 - physical,
 - chemical, or
 - microbiological origin
- The preparation of sterile –superior level of
 - The standards established,
 - the attitude of personnel, and
 - the process control.

➤ *Solvent*

- *Water for injection (WFI)*

➤ *Solubilizers*

- *for drugs that are poorly soluble in water*
 - *cosolvents*

➤ *Antimicrobial preservative Agents*

- required in preparations intended for multiple dosing from the same container

Cont...

➤ *Tonicity Adjusters*

- Reduce pain of injection

➤ Antioxidants

- Prevent oxidization by being oxidized faster than the drug or by blocking oxidization
- Water soluble-ascorbic acid
- Oil-soluble-butyl hydro toluene

➤ Buffers

- Added to maintain pH
- Results in stability

Pharmaceutical Additives Used in Parenteral Products

Function	Agent	Concentration (%)
Preservatives	Benzalkonium chloride	0.01
	Benzyl alcohol	2.0
	Chlorobutanol	0.5
	Chlorocresol	0.1–0.3
	Cresol	0.3–0.5
	Methyl paraben	0.18
	Propyl paraben	0.02
	Phenol	0.5
	Phenylmercuric nitrate	0.002
	Thimerosal	0.01
Antioxidants	Ascorbic acid	0.02–0.1
	Butyl hydroxyanisole	0.005–0.02
	Butyl hydroxytoluene	0.005–0.02
	Sodium bisulfite	0.1–0.15
	Sodium formaldehyde sulfoxylate	0.1–0.15
	Thiourea	0.005
	Tocopherol	0.05–0.075
Chelating agents	Ethylenediamine tetraacetic acid salts	0.01–0.075
Buffering agents	Acetic acid and a salt	1–2
	Citric acid and a salt	1–3
	Acid salts of phosphoric acid	0.8–2
Tonicity agents	Dextrose	5
	Sodium chloride	0.9

Rx *Methyldopate HCl Injection* (Aldomet[®], Merck)

Methyldopate hydrochloride: 250 mg

Methyl paraben: 7.5 mg

Propyl paraben: 1 mg

Sodium bisulfite: 16 mg

Monothioglycerol: 10 mg

Disodium edetate: 2.5 mg

Citric acid anhydrous: 25 mg

Sterile water for injection, USP *q.s.*: 5 ml

Methyldopate hydrochloride is an antihypertensive agent for i.v. use. Methyl and propyl parabens are added as preservatives, while sodium bisulfite and monothioglycerol are added as antioxidants. Disodium edetate serves as a chelating agent and citric acid as a buffering agent.

Large Volume parenteral

- Injections intended for intravenous use
- They are packaged in containers holding 100mL or more
- Other sterile large-volume solution includes those used for irrigation or dialysis.
- All LVPs are required to be:
 - Free of MOs
 - Non-pyrogenic
 - Free of particulate matter
 - The volume must not exceed 1000mL (except for irrigation solutions)



VisIV container



Intravenous solution packaged in pliable plastic.



Accurate delivery of intravenous fluids and medications by use of a controlled-rate infusion system (Alaris Medical Systems) for the drug vial and a volumetric

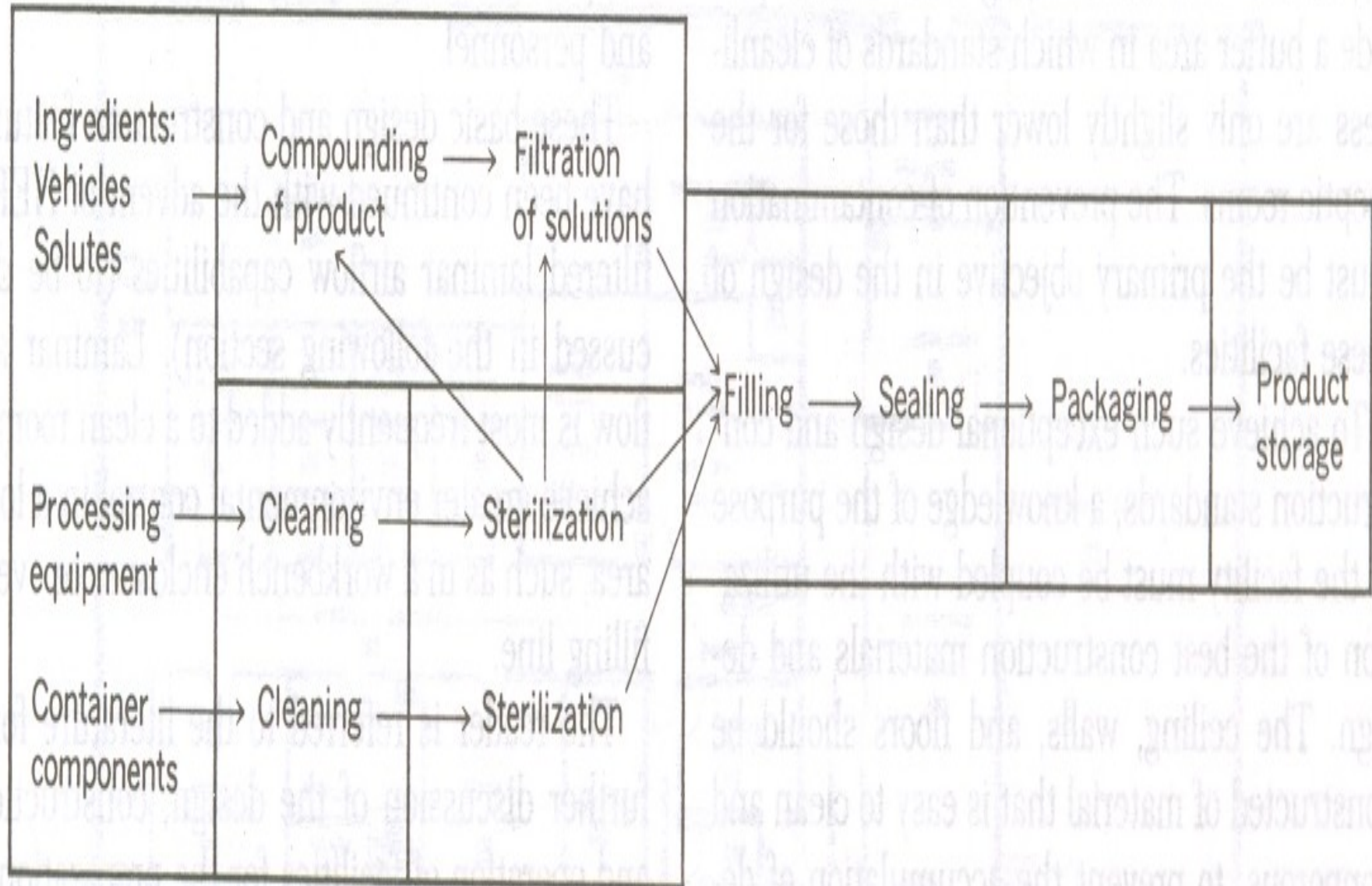
SOME INTRAVENOUS INFUSIONS THAT MAY BE ADMINISTERED IN VOLUMES OF 1 L, ALONE OR WITH OTHER DRUGS

INJECTION	USUAL CONTENTS	CATEGORY AND COMMENTS
Amino acid	3.5, 5, 5.5, 7, 8.5, and 10% crystalline amino acid with or without varying concentrations of electrolytes or glycerin	Fluid and nutrient replenisher
Dextrose injection, USP	2.5, 5, and 10% dextrose, other strengths	Fluid and nutrient replenisher
Dextrose and sodium chloride injection, USP	Dextrose 2.5%–10%; NaCl 0.11%–0.9% (19–154 mEq sodium)	Fluid, nutrient, electrolyte replenisher
Mannitol injection, USP	5, 10, 15, 20, and 25% mannitol	Diagnostic aid in renal function; diuretic; fluid and nutrient replenisher
Ringer's injection, USP	147 mEq sodium, 4 mEq potassium, 4.5 mEq calcium, and 156 mEq chloride per liter	Fluid and electrolyte replenisher
Lactated Ringer's injection, USP	2.7 mEq calcium, 4 mEq potassium, 130 mEq sodium, and 28 mEq lactate per liter	Systemic alkalinizer; fluid and electrolyte replenisher
Sodium chloride injection, USP	0.9% NaCl	Fluid and electrolyte replenisher; isotonic vehicle

Production

- The production process includes all of the steps from the accumulation and combining of the ingredient of the formula to the enclosing of the product individual container for distribution
 - To enhance the assurance of successful manufacturing operation, SOPs are important.
- passage of materials
 - formula ingredients, container components and processing equipment that have been released for use are drawn from their respective storage areas
 - compounded in an environment designed to maintain a high level of cleanliness.

Flow of materials through the production departments



Cont...

- Process equipment and container components
 - cleaned thoroughly according to required specifications,
 - are assembled in a clean environment,
 - are sterilized and
 - depyrogenated prior to use.
- All equipments and supplies introduced in to the aseptic filling area should be sterile coming directly from the sterilization process.
 - All supplies must be introduced into the aseptic filling rooms in such a manner that the aseptic state of these rooms is maintained
- The product is sealed in the final container within the aseptic room. it is then transferred to the packaging area
 - This area is maintained clean but need not meet the standard imposed for the aseptic rooms or for the compounding room.
- Packaged products are placed in quarantine storage until released by QC

Facilities

- should be designed for controlled cleanliness appropriate for each steps
 - near perfect cleanliness must be achieved in the aseptic filling rooms
 - surrounding area should provide a buffer area (only slightly lower cleanliness)
 - *prevention of continuation must be the primary objective in the design of these facilities*
- *construction materials*
 - The walls, floors should be constructed of material that is easy to clean and nonporous to prevent the accumulation of debris and moisture
 - sealing and walls
 - “spray on tile” - ceramic epoxy finish applied by spray or painting to form a continuous smooth seal coating

Cont...

- Floors
 - ceramic plastic cement
- Glass often used in partition to permit supervisory view
- furniture should be of non porous hard- surface materials
- Laminar air flow is most frequently added to a clean room
 - achieve greater environment controlling in a local area such as in a workbench enclosure or over a filling line.

A. Environment control

- The standard of environmental control varies depending on the area involved
 - clean-up,
 - packaging,
 - compounding or filling and
 - the type of product being prepared.
- product prepared aseptically (without terminal sterilization)
 - must be maintained under the most rigid control that existing technology permits.
- terminally sterilized product
 - some what less rigid biologic control of the compounding and filling area may be acceptable.
 - Daily uses of the disinfecting producers are usually expected for the clean up and packaging area.

B. Traffic control

- controlling and minimizing traffic particularly in and out of aseptic areas is essential
- The only access directly from the outside is
 - to the personnel wash rooms,
 - the equipment wash rooms,
 - the non sterile manufacturing area and
 - the capping (packaging) area
- access to the aseptic corridor, compounding and filling rooms
 - only through an airlock pass
 - permit controlled passage of supplies and personnel from non aseptic to aseptic areas

Cont...

- Personnel should be permitted to enter aseptic room only after following rigidly prescribed procedures
 - removing their street clothing,
 - washing their hands
 - wearing gowns, hats, shoes, facemasks, glove and other prescribed attire
- Once they have entered the aseptic area they should not be permitted to move in and out of the area without re-gowning.
- unauthorized personnel should never be permitted to enter the aseptic area

C. House keeping

- not one remaining particle of debris is acceptable.
- All equipment and surrounding work area must be cleaned thoroughly at the end of working day
 - No contaminating residues from the concluded process may remain on the ceiling, walls and other structures.

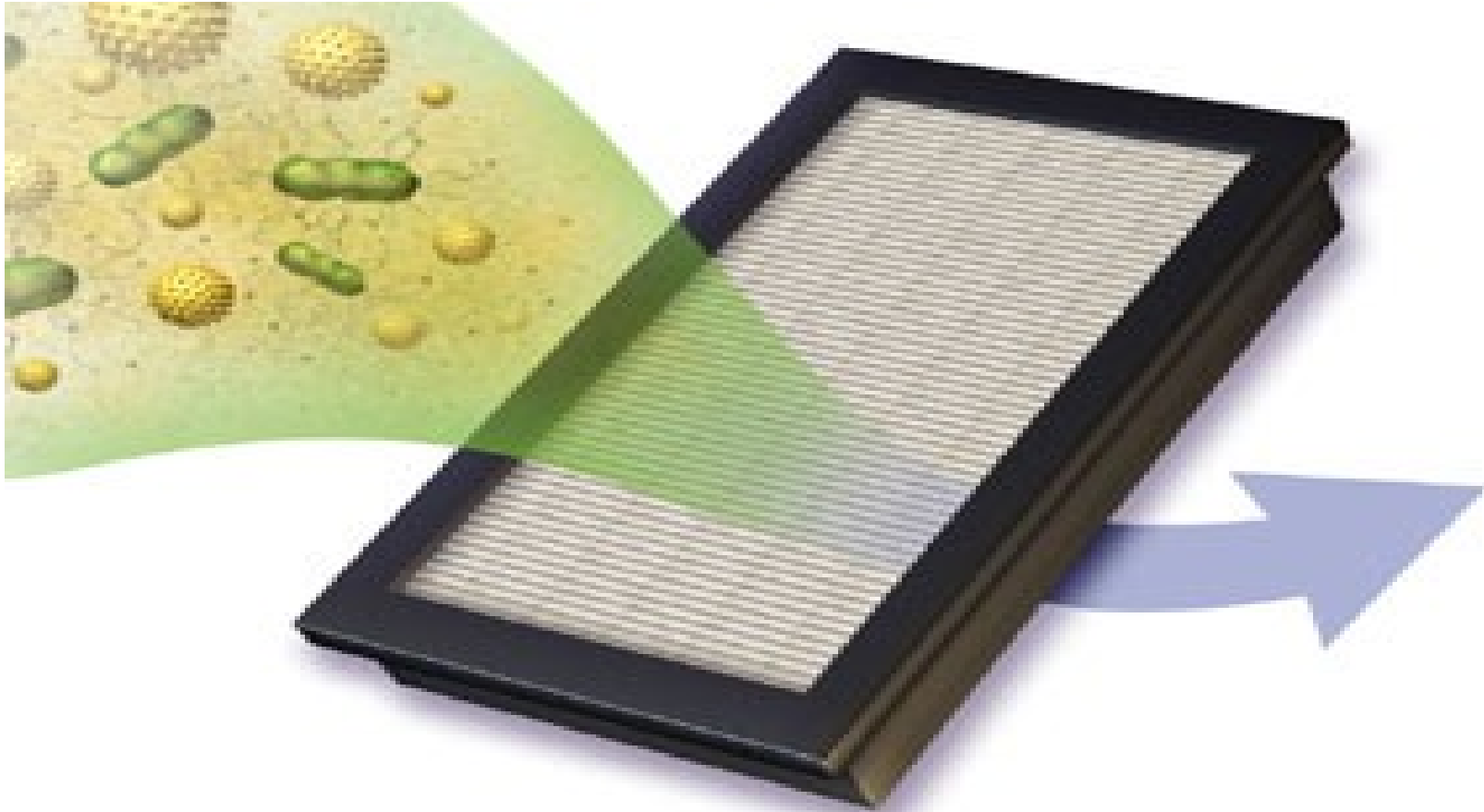
D. Surface disinfection

- After thorough cleaning all surfaces should be disinfected at least in the aseptic area
 - An effective liquid disinfectant should be sprayed on all surfaces
- ultraviolet lamps are located to provide adequate radiation.
 - reduces viable microorganisms present on surface area and in the air
 - useful to irradiate the inside exposed surface of
 - processing tanks,
 - surface under hoods,
 - the surface of conveyer belts and similar confined surface that are otherwise difficult to render aspects,
 - however, they cannot reach unexposed surface such as pipe connection.

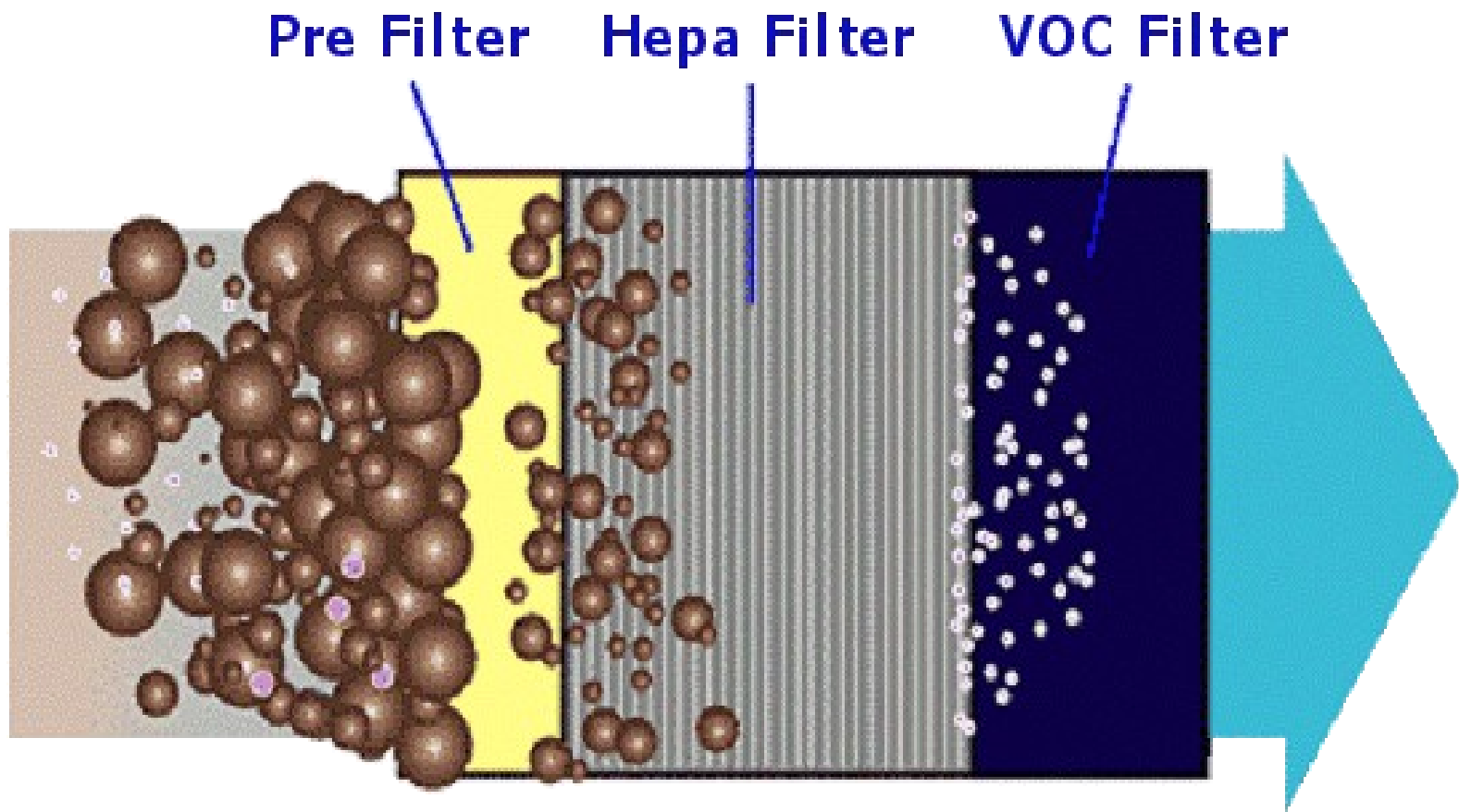
E. Air control

- In any area occupied by personnel the air must be exchanged at frequent intervals.
 - Fresh outside or recycled air must first be filtered to remove gross particulate matter.
- more than one pre filter may be used in series
 - the first of quite large and the next of somewhat smaller pore size to provide gradation of particle size removal from heavily contaminated air.
- To remove finer debris HEPA filters are employed
 - at least 99.97% efficient
 - remove particles of 0.3 micron size and larger and
 - composed of glass fibers and fillers or electrostatic precipitators

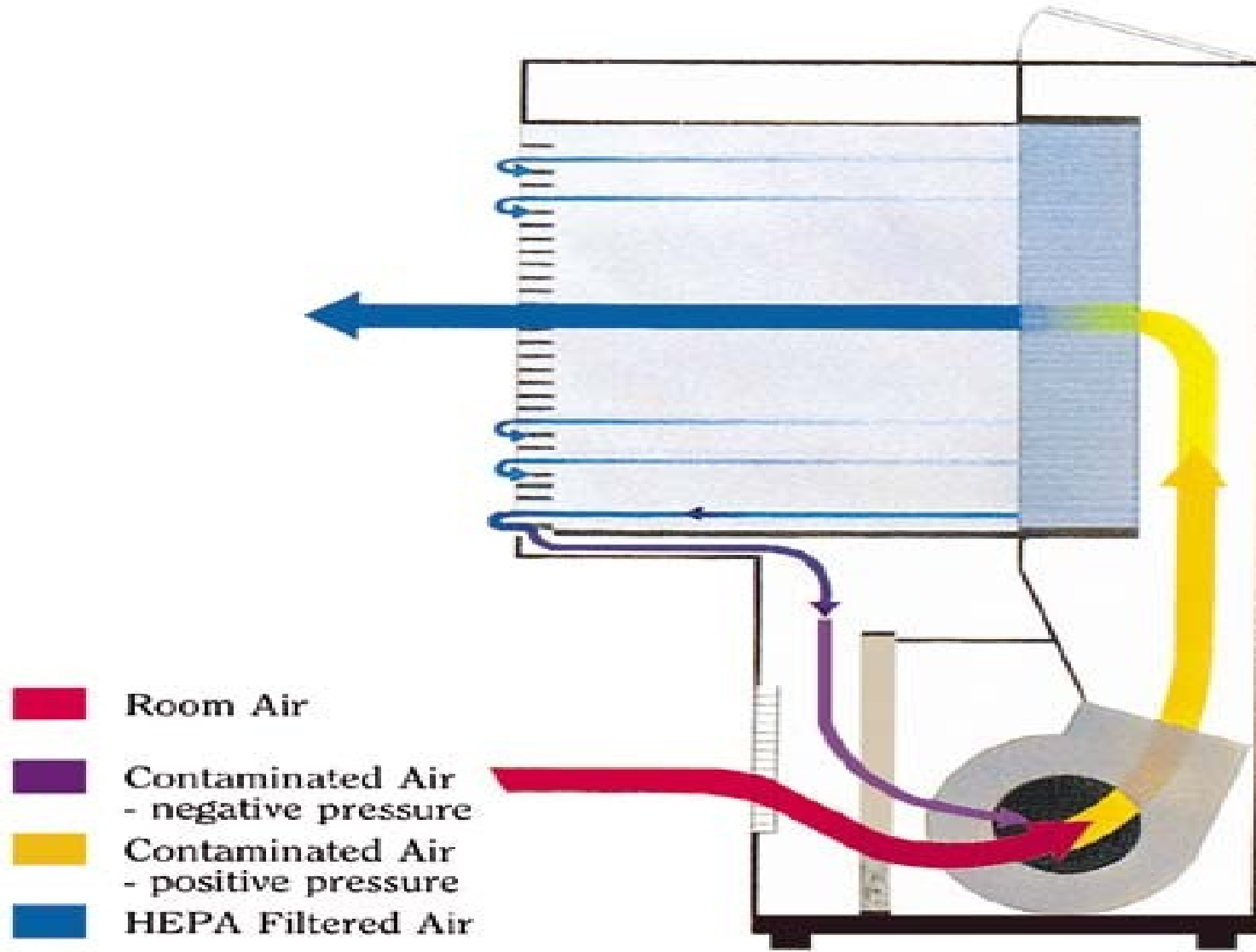
HEPA



Cont...



Horizontal laminar air flow



Laminar flow hood



Personnel

- The people who produce sterile products usually are nonprofessional persons supervised by those with professional training
 - they must be
 - inherently neat,
 - Orderly and reliable and
 - have good manual agility
- All employees should be in good health & should be subjected to periodic physical examinations
- They should report the developing symptoms of
 - a cold, a sore throat or other infectious diseases so that they can be assigned to less critical areas
- The clothing worn by personnel in the aseptic area usually consists of
 - sterile coveralls hoods,
 - face mask and shoe covers,
 - sterile rubber gloves also may be required

Cont...

- Personnel entering the aseptic area should be required to follow a definite preparatory procedure
 - removing at least outside street clothing,
 - scrubbing the hands and arms thoroughly with a disinfectant soap and
 - donning the prescribed uniform (Dacron cloth).
 - A full body water and soap shower
 - An air shower for the fully attired worker may be used at times to blow away loose lint
- Personnel working in equipment wash rooms, sterilizing room and packaging area
 - normally required to do clean uniform daily and to be conscious of cleanness

wearing

headgear

Beard cover and face mask

Gown

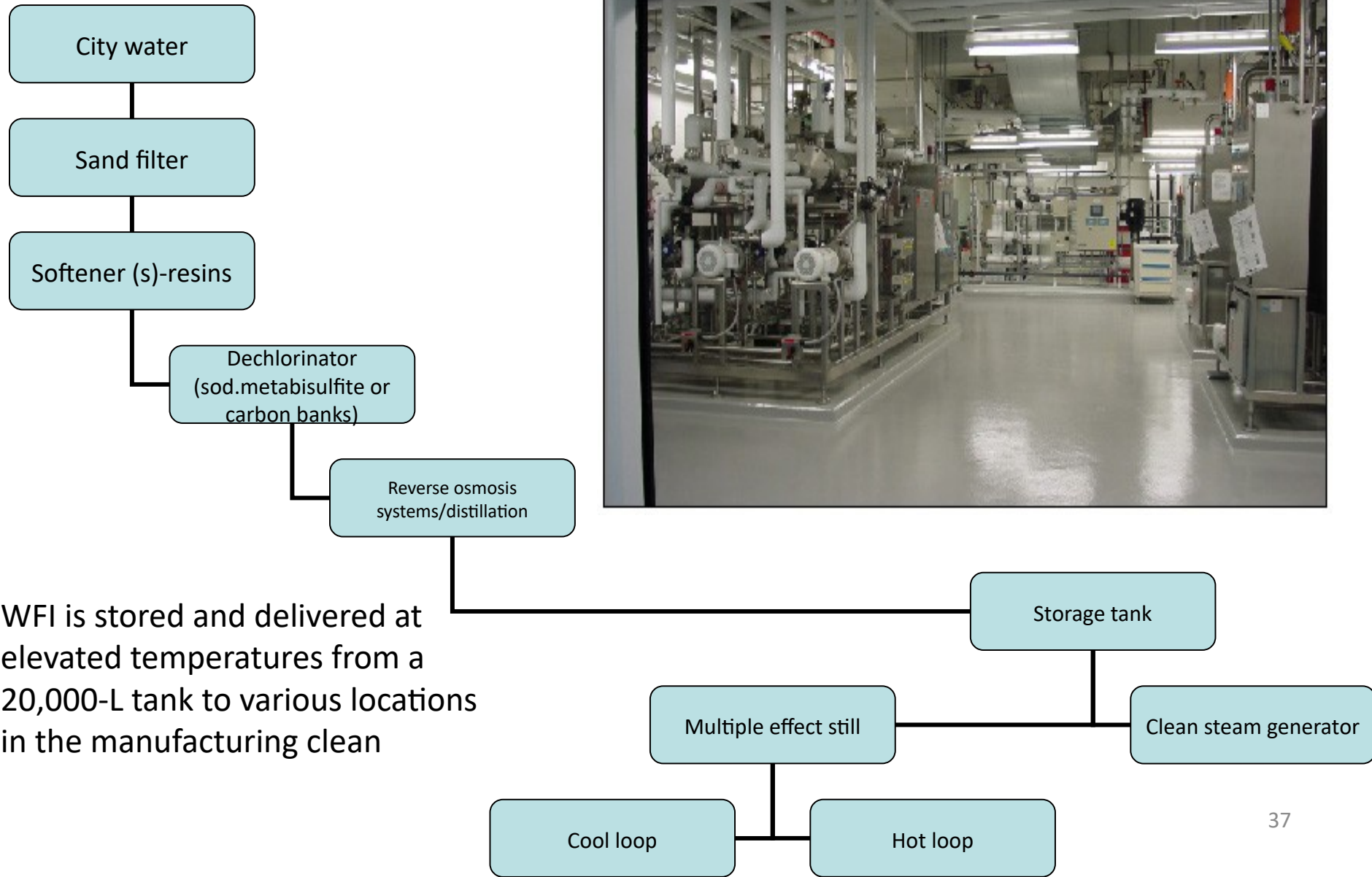
Gloves

Shoe covers



Water for injection (WFI)

prepared by distillation in a still or Reverse osmosis
steps from source to end



WFI is stored and delivered at elevated temperatures from a 20,000-L tank to various locations in the manufacturing clean

Cont...

- high-purity water used as a vehicle for injectable preparations which will be subsequently sterilized.
- can be stored for less than 24 hr at RT or for longer times at specific temperatures (80oC).
- needs to meet USP pyrogen test.
- maximum 1mg/100 ml total solids.
- may not contain an added substance.

Production...

- All equipment and supplies in filling area should be sterile
 - having come directly from the supplier
 - preferably through autoclaves,



Filtration

- Filtration is a unit operation process in which particles are separated from a liquid through a permeable material
- Bulk drug sterilization is accomplished by filtration, normally a depth filter.
 - The drugs can be pre-sterile filtered (e.g. $.45\ \mu\text{m}$), followed by a series of at least two sterile filters at $.22\ \mu\text{m}$.



Filling

- filling lines are designed for filling operations under aseptic and sterile conditions.

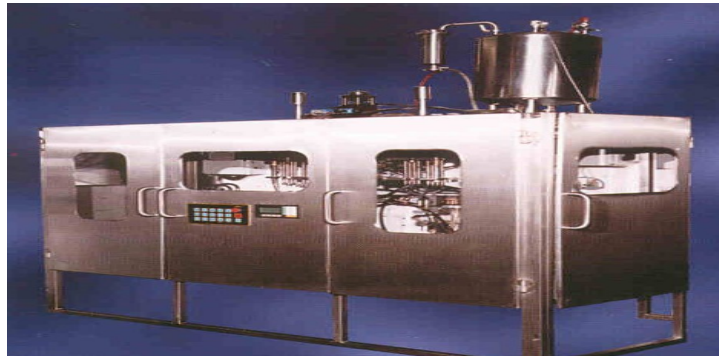


Filling...

- A measured volume of liquid through the orifice of a **delivery tube** designed to enter the constricted opening of a container.
- The size of the delivery tube is governed by
 - ▶ the opening in the container to be used,
 - ▶ the viscosity and density of the liquid, and
 - ▶ the speed of delivery desired.
- The tube must
 - freely enter the neck of the container and
 - deliver the liquid deep enough to permit air to escape

Blow/fill/seal technology

- Blow/fill/seal units are purpose built machines in which, in one continuous operation, containers are formed from a thermoplastic granulate, **filled** and then **sealed**, all by the one automatic machine
- The plastic used to make the container is normally low, medium or high-density **polyethylene** or **polypropylene**, depending on the desired **container properties**

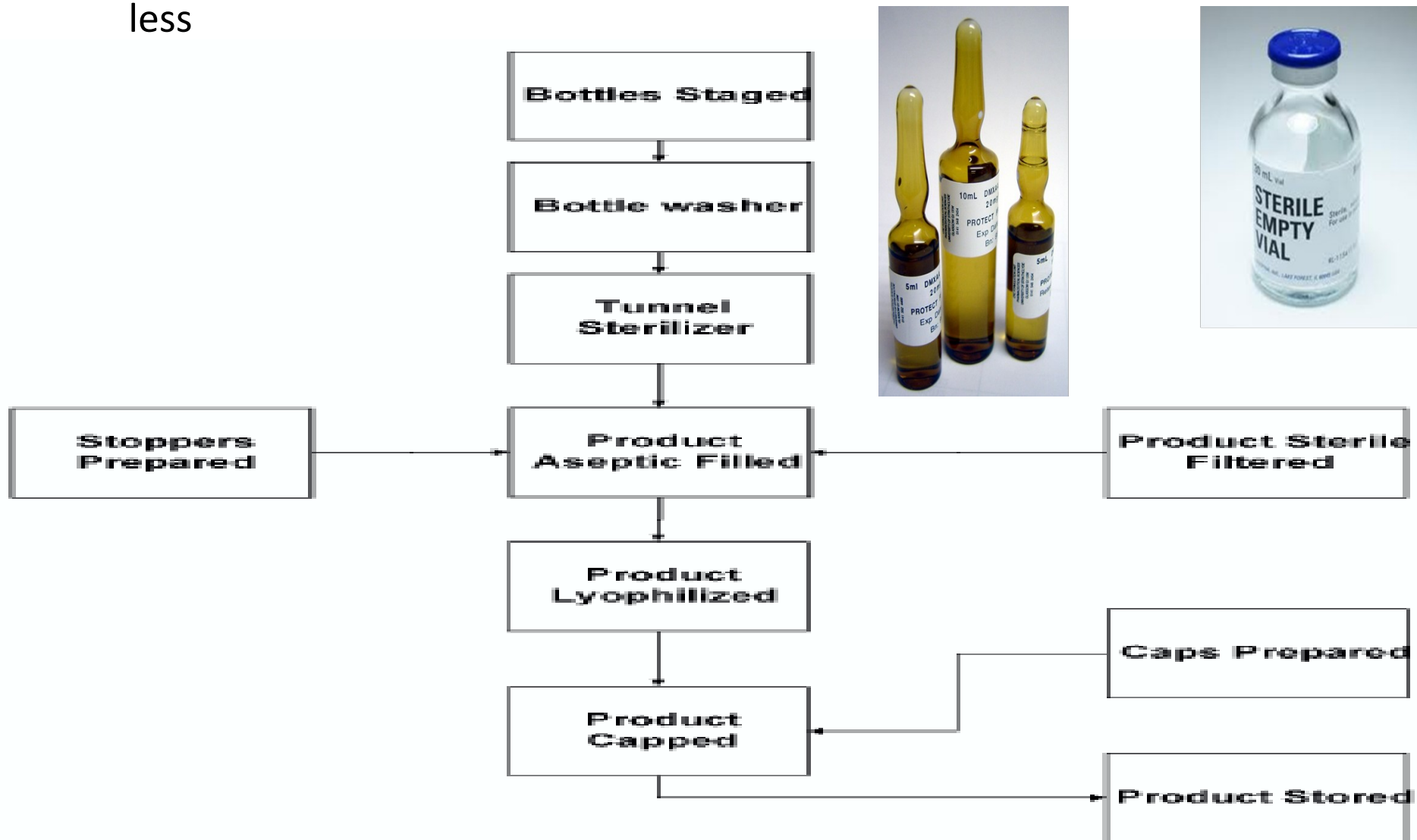


Advantages of BFS Technology

- **Cleaning** and **sterilizations** of prefabricated containers and closures is not required. A clean, sterile container is made within the BFS machine as it is required for filling
- The cost of material transport, storage and inventory control is reduced.
- **Validation** requirements are reduced
- A single compact BFS machine takes the place of several conventional machines, **saving floor space**
- In addition, zones for transport to successive filling and closing procedures are not required because these operations all take place in the BFS machine itself
- The operation of BFS machines is less **labor-intensive** than conventional aseptic filling

Small volume parenterals

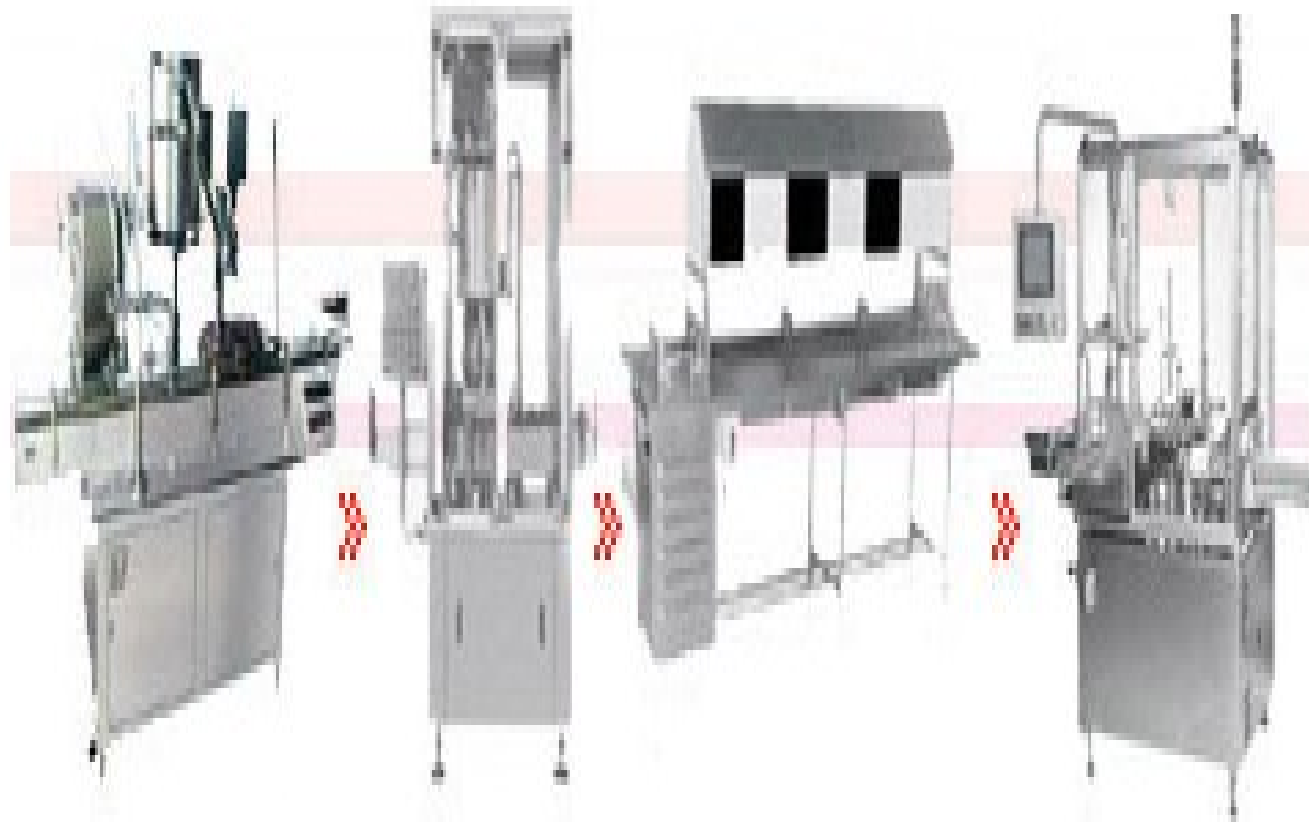
- Injection that is packaged in containers labeled as containing 100 mL or less



Cont...



WASHING



FILLING

SEALING

INSPECTION

LABELING

Techniques of Ampoules /vials wash

- The ampoules / vials can be fed pa
 - immersed in the heated ultras
- Pushed into and then fed directly c
 - to the jetting stations in the up
- Blown of and lifted into the discha
 - Trayed or fed directly on to the



Sterilization

- In case of steam sterilization up 30minutes



Liquid filling

- Product physical characteristics may suggest selection of one type of filling machine over another
- Low density liquid can be achieved
 - with light-duty machinery,
- Viscous , sticky, or high-density liquids
 - require much more rugged machines
 - to withstand the pressure required to dispense them.



Solid Filling

- are more difficult to subdivide accurately and precisely into individual dose containers than are liquids
 - rate of flow of solid material tends to be slow and irregular, particularly if finely powdered. Small, granular particles flow most evenly.
- major problems resulting in uneven filling
 - stratification of particles due to varying particles size,
 - the development of electrostatic charge
 - the formation of air pockets and
 - uneven flow due to clumping of the particles

Stoppering

- Stoppering of the vial will be done after filling
- Stopper size will need to match the finish of the selected vial.
- **Vibratory stopper bowl** is used to orient vial stoppers so that they can be fed into automated placement equipment.



Stoppering

- Compatibility of the product with the stopper's rubber formulation will dictate the stopper's physical characteristics.
 - A mineral-oil-based product will be incompatible with a butyl rubber stopper; a neoprene stopper is a better choice
- When filling of a low concentration product is required,
 - a fluoropolymer-coated stopper may be most suitable to reduce adsorbance of the product to the stopper's surface



Lyophilization

- Lyophilization, commonly referred to as freeze drying, is the **process of removing water from a product** by sublimation and desorption.
- This process is performed in Lyophilization equipment which consists of
 - ◆ a drying chamber with temperature controlled shelves,
 - ◆ a condenser
 - ◆ a cooling system
 - ◆ a vacuum system
- For pharmaceutical compounds that undergo **hydrolytic degradation**, Lyophilization offers a means of improving their **stability** and shelf life

Lyophilization...

- Many parenteral medications such as vaccines, proteins, peptides, and antibiotics have been successfully lyophilized
- Lyophilization cycles consist of three phases
 - Freezing,
 - primary drying, and
 - secondary drying
- During the freezing phase, the goal is to freeze the mobile water of the product
- The **rate of cooling** will influence the **structure of the frozen matrix**

Lyophilization equipment



SEALING

- Containers should be sealed in the aseptic area immediately adjacent to the filling machine
- a sterile container that has been opened no longer be considered to be sterile.
 - tamper proof sealing is essential.
- Ampoules
 - bead seals (tip-seals) or pull-seals

Sealing vials

- Rubber closures must fit the opening of the container snugly enough to produce a seal
 - Aluminum caps are used to hold rubber closure in places

Quality control

Pyrogen tests

- traditional test uses rabbits, solution injected into the ear vein using sterile device
 - measure body temperature
 - a temperature elevation will occur within 3 hours
 - Many imitative medical agents will also cause a fever
- limulus amebocyte lysate(LAL) tests
 - ✱ A test sample is incubated with amebocyte lysate from the blood of the horseshoe crab, Limulus polyphemus.
 - ✱ A pyrogenic substance will cause a gel to form
 - ⇒clottable protein from the amebocyte cells reacting with the endotoxins
 - ✱ This test is more sensitive, more rapid, and easier to perform than the rabbit test

Leaker Test

- Capillary pores or tiny cracks
 - microorganisms or other dangerous contaminants may enter the ampoule, or
 - the contents may leak to the outside and spoil the appearance of the package
- Tip-sealed ampoules are more likely to be incompletely sealed than are those that have been pull-sealed
- usually in a vacuum chamber (27 inches Hg or more)
 - submerged ampoules in a deeply colored dye solution (usually 0.5-10% methyleneblue)
 - Subsequent atmospheric pressure then causes the dye to penetrate an opening
 - visible after the ampoule has been washed externally to clear it of dye

Clarity Test

- every unit of product is perfectly free from visible particulate matters
 - it is the responsibility of the quality control department to detect and discard individual containers of product that the ultimate user would consider unclear

packaging

- Packaging is an essential part of the parenteral formulation.
- Providing long-term protection and maintenance of physical and chemical stability
- The main types of packaging systems
 - glass,
 - rubber, and
 - plastic

Terminal Sterilization/Depyrogenation

- Sterilization - is the process designed to produce a sterile state.
- is the absolute condition of total destruction or elimination of all living microorganisms
- Aseptic - indicates a controlled of process or condition in which the level of microbial contamination is reduced to the degree that microorganisms can be excluded from a product during processing.

Types of Sterilization

There are two basic types of sterilization methods

- Physical sterilization and
- Chemical sterilization

1. Physical processes of sterilization

There are two methods of thermal sterilization

a. Dry heat

Used for substances that resist degradation above approximately 140°C (284°F).

- 170°C for 1 hour

- 160°C for 2 hours

- 140°C for 4 hours.

or 45 min at 260°C (500 °F) normally can be expected to kill spores as well as vegetative forms of all microorganisms.

- oils and other aqueous vehicles (e.g. glycerin, propylene glycol)

- heat-stable therapeutic agents/excipients

- glassware (e.g. bottles).

- The time to produce a lethal effect is inversely proportional to the temperature employed.

b. Moist heat

- Moist heat is most effective than dry heat for thermal sterilization because moist heat causes the coagulation of cell protein at a much lower temperature than dry heat and also the thermal capacity of steam is much greater than that of hot air.
- for the sterilization of aqueous solutions, glassware, and rubber articles.
- It is not suitable for sterilizing solutions of drugs that are thermolabile or are liable to degrade in the presence of moisture.
- Oily or fatty substances and powders cannot be sterilized by this means.

- used to sterilize materials that are both thermostable (within the conditions of the sterilisation cycle) and through which moisture can perfuse. These include:
 - glassware
 - dressings
 - closures
 - aqueous solutions:

Non thermal methods

1. Ultraviolet light

- Aid in the reduction of contamination in the air and on surfaces within the processing environment.
- The germicidal light produced by mercury vapor lamps is emitted almost exclusively at a wavelength of 2537 angstrom units (253.7 μm).

2) Ionizing radiations

- They are high energy radiations emitted from radioactive isotopes such as cobalt.
- γ (gamma rays) or produced by mechanical acceleration of electrons to very velocities and energies (cathode rays, beta rays).

- Thermolabile drugs such as penicillin, streptomycin, thiamine, and riboflavin have been effectively sterilized by ionizing radiation.
- Pharmaceutical products are more resistant to degradation by ionizing radiation if they are in a powdered form than in a liquid form.
- Insulin and heparin solutions are inactivated by the radiation.

2. Chemical processes of sterilization

a) Gas sterilization

- These gases are highly reactive chemicals, however, and are difficult to remove from many materials after exposure.
- Example:- Formaldehyde and sulfur dioxide (Their usefulness is limited)
- There are two newer gases with fewer disadvantages and used for their sterilization importance.

Example:- Ethylene oxide and beta - propiolactone.

- Ethylene oxide sterilization is effective in sterilizing a wide variety of materials including surgical instruments and gloves, plastic syringes, disposable needles, tubing sets, and dialysis units.
- In addition, it is often used to sterilize thermolabile powdered drugs such as penicillins.
- However, some drugs, for example, thiamine, riboflavin, and streptomycin, lose potency when treated with ethylene oxide.



Thank you